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Key Points:
- The Coldwell Complex records only a single symmetrical geomagnetic reversal
- The reversal asymmetry at ~1.1 Ga reflects a fast North American plate motion
- The fast plate motion may reflect a decreased mantle drag due to unusually vigorous mantle

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Paleomagnetism of the ~1.1 Ga Coldwell Complex (Ontario, Canada): Implications for Proterozoic geomagnetic field morphology and plate velocities

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Abstract We report new paleomagnetic data from the ~1108 Ma intrusive Coldwell Complex (Ontario, Canada) to investigate the apparent reversal asymmetry observed in some Midcontinent Rift (MCR) rocks. The rocks of eastern and central part of the complex are reversely magnetized with a group mean direction of $D = 114.8°$, $I = -63.7°$ ($\alpha_{95} = 3.6°$, $N = 30$). The corresponding paleomagnetic pole at $P_{lat} = 47.2°$N, $P_{long} = 206.5°$E ($A_{95} = 4.8°$) is located close to the paleomagnetic poles from nearly coeval reversely magnetized rocks of the MCR system, including the lower lava flows at Mamainse Point. The rocks of western part of the complex are normally magnetized with a group mean direction ($D = 298.0°$, $I = 56.9°$, $\alpha_{95} = 5.8°$, $N = 10$) that passes the reversal test with respect to the reversed group mean direction. Our results do not support the previous model in which the complex was emplaced during two periods of reversed geomagnetic field polarity separated by a period of normal polarity and hence encompasses two geomagnetic reversals. Instead, our new data indicate that the Coldwell Complex records only two polarity intervals separated by a symmetrical reversal at ~1102–1105 Ma. This reversal is likely equivalent to the lowermost reversal recorded at Mamainse Point and provides further evidence that the apparent reversal asymmetry reflects a plate motion rather than a persistent nondipole field geometry. Together with a high-quality data from the ~1098 Ma North Shore Volcanics, our data indicate a rapid velocity of Laurentia at ~25 ± 4 cm/yr. The fast plate motion may reflect a decreased mantle drag due to vigorous mantle indicated by widespread intraplate magmatism at ~1.1 Ga.

1. Introduction

The geocentric axial dipole (GAD) assumption, a keystone of paleomagnetic research, states that the Earth's magnetic field when averaged over a sufficient amount of time represents the field of a geocentric dipole aligned with the Earth's spin axis. The GAD assumption implies the symmetry of geomagnetic reversals which, in the paleomagnetic context, is expressed as an exact antiparallelism of the directions of time-averaged field before and after a reversal. All known geomagnetic reversals are symmetrical throughout the Neoarchean [e.g., *Jacobs*, 1994; *Strik et al.*, 2003] thus confirming the long-term validity of GAD assumption. The only possible exception is an apparent geomagnetic reversal asymmetry manifested in rocks of the ~1.1 Ga North American Midcontinent Rift (MCR) system that crop out around Lake Superior [e.g., *Palmer*, 1970; *Halls and Pesonen*, 1982; *Schmidt and Williams*, 2003]. Most of the MCR-related lava flows and intrusive rocks that are reversely (upward inclination) magnetized consistently have characteristic directions of remanent magnetization that are about 15–30° steeper in magnetic inclination than their normally (downward inclination) magnetized equivalents, while declinations show the expected 180° relationship.

Until recently, the two most favored hypotheses for this reversal asymmetry were either apparent polar wander during the time of MCR emplacement [*Davis and Green*, 1997; *Schmidt and Williams*, 2003; *Swanson-Hysell et al.*, 2009] or the presence of a persistent nonaxial dipole field causing the geomagnetic field to depart from a geocentric axial dipole geometry [*Pesonen and Nevanlinna*, 1981; *Halls and Pesonen*, 1982; *Nevanlinna and Pesonen*, 1983].

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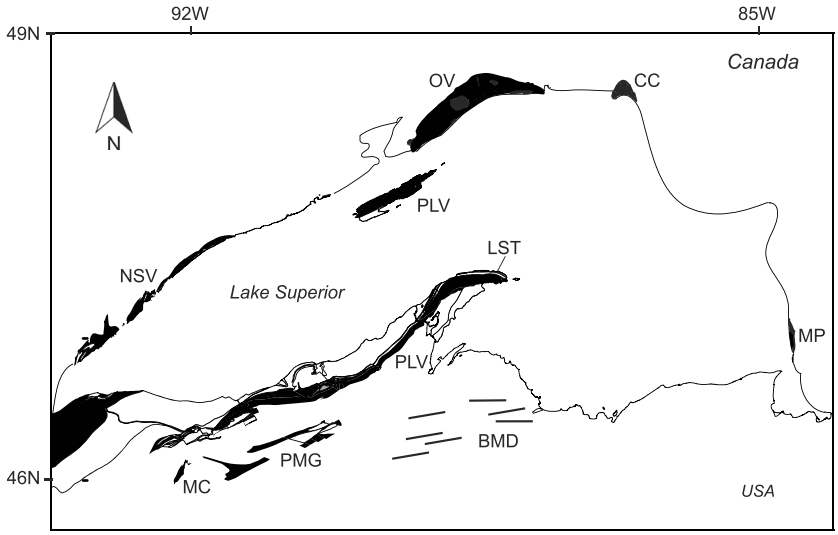


Figure 1. Location of the Coldwell Complex (CC) and other selected rock sequences of the Midcontinent Rift system: MP, Mamainse Point lava flows; OV, Osler Volcanics; NSV, North Shore Volcanics; MC, Mellen Complex; PMG, Powder Mill Group; PLV; Portage Lake Volcanics; LST, Lake Shore Traps; and BMD, Baraga-Marquette dikes.

Unfortunately, direct testing of the reversal asymmetry is impossible because of the lack of known MCR sequences that record transitional field directions between the two stable asymmetric directions. However, there are two MCR locations that presumably record multiple reversals which makes them crucial for evaluating the reversal asymmetry: the Mamainse Point volcanics [*Palmer*, 1970; *Robertson and Fahrig*, 1971; *Robertson*, 1973; *Swanson-Hysell et al*., 2009] and the intrusive Coldwell Complex [*Lewchuk and Symons*, 1990], both located in Ontario, Canada.

The basaltic flows of the Mamainse Point sequence record four polarity intervals: lower reversed (MR1), lower normal (MN1), upper reversed (MR2), and upper normal (MN2) [e.g., *Robertson*, 1973]. A recent paleomagnetic investigation [*Swanson-Hysell et al*., 2009] showed a gradual shallowing of inclination from the base to the top of the sequence reflecting a North America plate motion toward the equator. The authors obtained positive reversal tests for all three Mamainse Point reversals and concluded that the reversal asymmetry is a result of an apparent polar wander that appears because other MCR sequences do not contain any record of the MN1 and MR2 polarity zones. We note, however, that these reversal tests were based on a limited number of paleomagnetic site mean directions, most likely insufficient to average out the paleosecular variation of geomagnetic field. Most notably, the test for the lowermost (oldest) MR1 → MN1 reversal utilized only six directions, one of which was based on only two specimens. If this site mean direction is excluded, the reversal test becomes indeterminate. Therefore, a possibility of reversal asymmetry remains, warranting additional tests, especially for the MR1 → MN1 reversal.

The paleomagnetism of the Coldwell Complex was previously investigated by *Lewchuk and Symons* [1990] who argued that the complex was emplaced in three magmatic episodes during two periods of reversed field polarity (CR1 and CR2) separated by a period of normal polarity (CN), hence recording two geomagnetic reversals (CR1 → CN and CN → CR2). Based on their interpretation of paleomagnetic data, the authors concluded that both reversals were symmetrical. They also proposed a model in which the central part of the complex represented the latest magmatic episode that followed the emplacement of the normally magnetized rocks of the western part, a proposition that contradicts the earlier complex emplacement models based on the geological field relationships [e.g., *Mitchell and Platt*, 1978]. However, as we discuss below, on closer examination, the data presented by *Lewchuk and Symons* [1990] do not provide an unambiguous answer with respect to the reversal asymmetry and to the relative age of magmatic episodes. This ambiguity motivated our paleomagnetic reinvestigation of the Coldwell Complex presented in this paper. Below, we report our new paleomagnetic data and discuss their implications for the reversal asymmetry as well as for the geological evolution of the complex.

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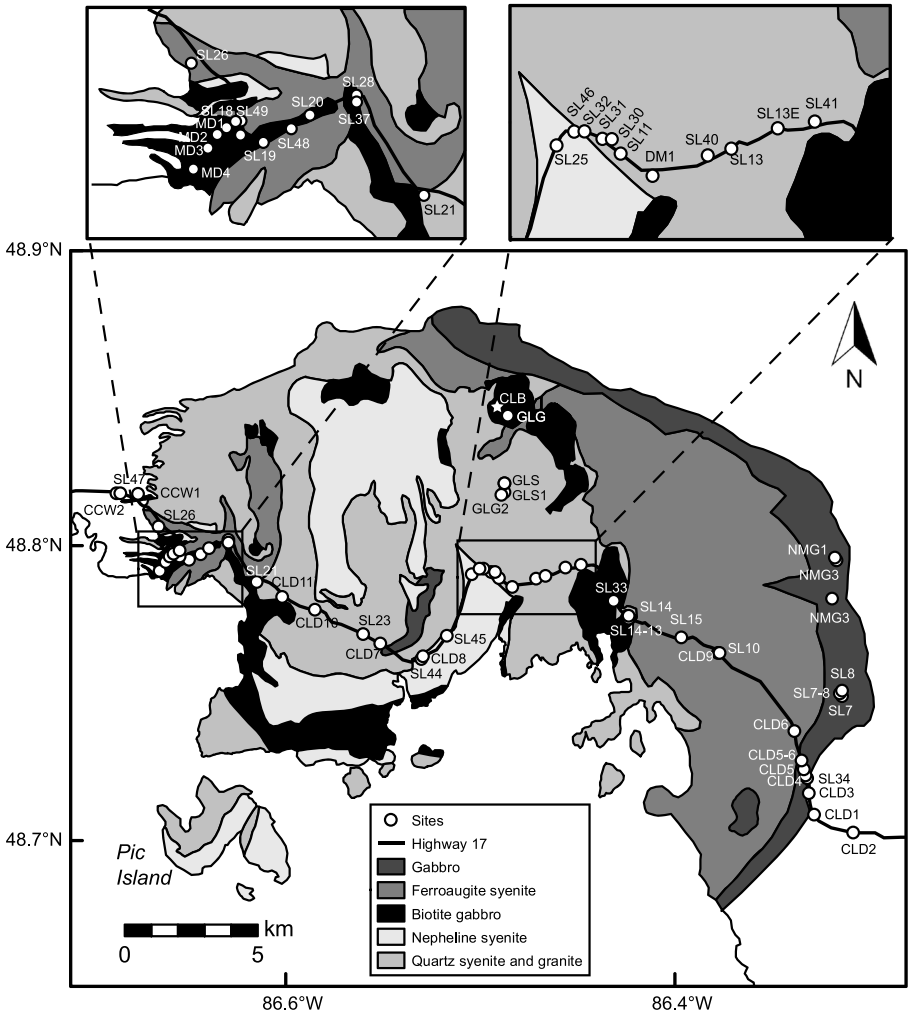


Figure 2. Simplified geology of the Coldwell Complex after *Puskas* [1967], *Platt and Mitchell* [1982], and *Currie* [1980]. Open circles show the locations of this study sampling sites.

2. Geology, Age, and Paleomagnetism of the Coldwell Complex

The multiphase alkaline Coldwell Complex occupies more than 350 km^2 on the north shore of Lake Superior near the town of Marathon (Ontario, Canada) (Figure 1) [e.g., *Sage*, 1991; *Sutcliffe*, 1991]. The complex is a subcircular composite intrusion approximately 25 km in diameter that intrudes Archean supracrustal and granitic rocks of the Schreiber-White Lake subprovince of the Superior province in Canada [e.g., *Puskas*, 1970; *Mitchell and Platt*, 1978, 1982; *Mitchell et al.*, 1993].

The eastern part of the complex consists of two main phases, massive gabbro along the eastern rim and the more abundant ferroaugite syenite [e.g., *Currie*, 1980] (Figure 2). The central part of the complex is chemically different from its eastern part and is composed primarily of biotite gabbros and nepheline syenites cut by several diabase dikes [e.g., *Mitchell and Platt*, 1982]. A few minor exposures of pillow basalts are also found in this area. In addition, a sequence of flat-lying basaltic flows crops out in the Coubran and Geordie lakes area [*Kulakov et al.*, 2012] (Figure 2). The western part of the complex is mainly composed of quartz syenites, syenites, and granites with large xenoliths of assimilated country rocks. Minor gabbro exposures are also present. The western rock units appear to be intrusive into most other phases of the complex [*Mitchell and Platt*, 1978]. No postintrusion metamorphism of the Coldwell rocks has been reported. Postintrusive structures are limited to block faults with minor offsets associated with regional uplift [*Mitchell and Platt*, 1982].

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Early geological studies considered the Coldwell Complex as a single intrusion that underwent extensive fractional crystallization of magma [*Lilley*, 1964; *Puskas*, 1967]. However, subsequent geochemical, petrological, and field relationship analyses resulted in a model in which the complex was emplaced in three magmatic episodes separated by quiescent periods as the magma source deepened and migrated to the western side of the complex [e.g., *Currie*, 1980; *Mitchell et al*., 1993; *Lewchuk and Symons*, 1990; *Walker et al*., 1993]. However, many details of the complex's geological history such as the extent and order of the intrusive episodes have been debated [e.g., *Currie*, 1980; *Mitchell and Platt*, 1978, 1982; *Lewchuk and Symons*, 1990].

According to *Currie* [1980], the eastern gabbro and ferroaugite syenites represent the earliest intrusive episode (Episode I). In his model, all the remaining rocks of the central and western part were emplaced during Episode II except for the granites and red syenites that were interpreted as metasomatized roof rock. The syenites on Pic Island were emplaced during the Episode III [*Currie*, 1980]. Alternatively, *Mitchell and Platt* [1982] argued that the western gabbros were also emplaced during Episode I, while Episode II was limited to the biotite gabbro and the nepheline syenite in the central part. In their model, the syenites, quartz syenites, and granites of the western part were emplaced during Episode III.

Early geochronological studies of the complex resulted in a wide range of Rb-Sr and K-Ar ages from 1335 to 1005 Ma, but the emplacement age was generally considered to be between 1070 and 1040 Ma [e.g., *Bell and Blenkinsop*, 1980; *Platt and Mitchell*, 1982], thus implying that the Coldwell Complex magmatism occurred late in the development of the rift, progressing from tholeiitic to alkaline with time [e.g., *Mitchell et al*., 1993]. On the other hand, *Turek et al*. [1985], based on a relatively imprecise U-Pb zircon date of 1188 ± 56 Ma from nepheline syenite of the central part, suggested that the complex was emplaced much earlier and experienced a prolonged cooling period in excess of 150 Ma. However, the validity of the age determination by *Turek et al*. [1985] has been contested [*Thorpe*, 1986].

More recently, in order to determine the emplacement ages of the intrusive centers, *Heaman and Machado* [1992] conducted a U-Pb zircon/baddeleyite geochronological study of the Coldwell Complex on two samples of the eastern gabbro, two samples of syenite and nepheline syenite from the central part, and a sample of granite from the western part. The gabbro samples yielded the most precise dates of 1108 ± 1 Ma and 1107 + 5/−1 Ma. For the nepheline syenite, a discordia line defined a similar upper intercept age of 1109 + 8/−4 Ma. For the syenite sample, a discordia line was not well defined but the majority of zircon analyses were consistent with the emplacement age of 1108 Ma. The four analyzed zircon fractions from the granite sample yielded $^{207}Pb/^{206}Pb$ ratios corresponding to model ages between 1103 and 1090 Ma but did not define a simple discordia line. Although *Heaman and Machado* [1992] assigned a 1107 + 9/−5 Ma age to this sample, their data do not rule out a possibility that this sample can be younger by a few millions of years.

Notwithstanding the deficiencies of their U-Pb analyses, the authors concluded that most of the Coldwell Complex was emplaced into "cold" Archean crust at 1108 ± 1 Ma and experienced a relatively rapid (< 3 Ma) cooling history [*Heaman and Machado*, 1992]. This age is similar to the U-Pb ages obtained for several other early MCR sequences, including the Logan Sills (1109 + 4/−2 Ma), a rhyolite porphyry from the base of the Osler Volcanic group (1108 + 4/−2 Ma) [*Davis and Sutcliffe*, 1985], and the 1107.3 ± 1.6 Ma Powder Mill basalts [*Davis and Green*, 1997]. Overall, however, the precision of available geochronological data from the Coldwell Complex is not sufficient to reliably resolve its emplacement history.

In the absence of reliable radiometric ages, paleomagnetism can provide time constraints on the number and order of intrusive episodes. The only published paleomagnetic investigation of the Coldwell Complex by *Lewchuk and Symons* [1990] that also incorporated an unpublished paleomagnetic data set obtained in 1972 by William A. Robertson identified three distinct directions (named A, B, and C) of the primary characteristic remanent magnetization (ChRM). The reversed polarity direction A with the mean of $D = 121.0°$, $I = -70.9°$ ($\alpha_{95} = 3.3°$, $N = 19$) was exclusively observed from the eastern gabbro and ferroaugite syenite. The normal polarity direction B with the mean of $D = 301.8°$, $I = 60.1°$ ($\alpha_{95} = 5.8°$, $N = 11$) was observed from the western gabbro and syenite. Finally, the reversed polarity direction C with the mean of $D = 119.1°$, $I = -54.3°$ ($\alpha_{95} = 5.1°$, $N = 9$) was obtained from the biotite gabbros, nepheline syenites, and syenites of the central part of the complex.

Because the reversed mean directions A and C were statistically different, *Lewchuk and Symons* [1990] concluded that the three ChRM directions corresponded to three separate intrusive episodes during two

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Table 1. Summary of Rock Magnetic Properties of the Coldwell Complex Samples[a]

Site	Lithology	NRM Range (A/m)	H_c (mT)	H_{cr} (mT)	M_{rs}/M_s
		Center A			
CLD1	Rhyolite	0.018–0.355	4.77	14.22	0.058
CLD3	E. gabbro	0.069–3.211	17.43	33.4	0.210
CLD4	E. gabbro	0.074–28.80	7.99	18.44	0.071
CLD5	E. gabbro	0.122–1.13	9.43	22.34	0.086
CLD6	Aug. syenite	0.104–0.271	8.66	24.65	0.072
NMG1	E. gabbro	0.305–45.88	6.38	18.2	0.057
SL10	Aug. syenite	0.044–0.184	4.56	11.7	0.059
SL14	Aug. syenite	0.0034–0.396	7.17	14.86	0.087
SL15	Aug. syenite	0.146–0.643	6.78	16.25	0.072
SL33	Basalt	2.305–21.58	14.31	22.59	0.183
SL34	Aug. syenite	0.591–36.43	7.32	16.94	0.043
SL5-6	E. gabbro	6.12–29.90	13.56	24.4	0.168
SL7	E. gabbro	1.621–2.418	7.51	13.07	0.097
SL7-8	E. gabbro	0.607–3.06	7.88	29.06	0.091
CLD2[b]	E. gabbro	0.0009–0.0019	16.6	23.18	0.252
CLD9[b]	Aug. syenite	0.405–0.757	11.43	19.32	0.237
SL8[b]	E. gabbro	5.95–10.11	1.39	4.24	0.013
NMG2[b]	E. gabbro	0.031–2.586	11.56	27.63	0.110
NMG3[b]	E. gabbro	0.36–1.98	3.38	13.25	0.036
		Center B			
CCW1	W. gabbro	0.004–0.098	10.92	13.44	0.131
CCW2	Pillowed lava	0.002–0.024	10.23	17.26	0.154
CLD7	Mafic dyke	0.005–0.229	4.87	15.89	0.207
SL13E	Mafic dyke	0.977–13.25	9.92	19.21	0.112
SL18	W. gabbro	4.748–30.06	4.93	9.38	0.101
SL19	W. gabbro	0.314–2.00	5.05	10.46	0.051
SL20	W. gabbro	0.604–59.47	7.33	12.43	0.124
SL37	W. gabbro	1.326–33.72	3.44	6.93	0.047
SL47	Pillowed lava	0.834–9.591	9.54	23.10	0.154
SL49	W. gabbro	6.864–88.68	4.75	28.56	0.025
MD1[b]	Gabbro	0.564–7.24	4.51	8.94	0.058
MD2[b]	Gabbro	0.03–30.12	3.00	7.84	0.034
MD3[b]	Gabbro	6.32–70.01	5.83	13.52	0.019
MD4[b]	Gabbro	0.777–2.826	7.45	14.86	0.095
SL26[b]	Aug. syenite	0.0001–0.343	10.55	20.83	0.114
SL48[b]	Neph. syenite	0.061–1.993	9.43	23.14	0.097
		Center C			
CLD11	Q-tz syenite	0.017–0.04	1.88	5.09	0.016
GLG	Gabbro	4.787–34.15	6.57	16.29	0.074
GLS	Q-tz syenite	0.097–0.438	4.36	10.12	0.051
GLS1	Q-tz syenite	0.01–0.067	5.18	11.49	0.042
CLD8	Neph.syenite	0.102–0.994	2.42	7.45	0.044
SL11	Amph. syenite	0.026–1.11	3.94	8.01	0.051
SL13	Amph. syenite	0.043–0.391	4.53	13.61	0.099
SL14-13	Augite syenite	0.665–5.797	6.99	13.77	0.082
SL21	Q-tz syenite	0.059–1.511	8.75	33.2	0.062
SL25	Neph. syenite	0.066–0.196	2.18	5.72	0.023
SL28	Amph. syenite	0.021–2.724	2.9	13.43	0.026
SL32	Diabase lava	0.012–9.104	8.27	17.9	0.145
SL41	Amph. syenite	0.025–1.362	2.47	7.13	0.023
SL44	Neph. syenite	0.148–3.857	3.57	10.8	0.034
SL45	Neph. syenite	0.184–1.59	4.89	16.11	0.053
SL46	Neph. syenite	0.081–9.537	1.38	4.91	0.013
CLD10[b]	Q-tz syenite	0.012–0.412	10.43	36.4	0.096
GLG2[b]	Gabbro	4.291–18.15	7.32	17.26	0.061
DM1[b]	Amph. syenite	2.98–96.63	7.55	18.43	0.1045
SL23[b]	Q-tz syenite	0.038–0.261	5.58	26.44	0.059
SL30[b]	Amph. syenite	0.118–1.012	7.57	16.94	0.108

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Table 1. (continued)

Site	Lithology	NRM Range (A/m)	H_c (mT)	H_{cr} (mT)	M_{rs}/M_s
SL31[b]	Diabase Lava	0.011–0.632	10.86	18.6	0.156
SL40[b]	Q-tz syenite	0.031–0.781	3.72	9.51	0.039

[a]NRM, natural remanent magnetization; H_c, coercivity; H_{cr}, coercivity of remanence; M_{rs}, saturation remanence; and M_s, saturation magnetization.
[b]Sites rejected from paleomagnetic mean calculation.

periods of reversed geomagnetic field polarity (directions A and C) separated by a period of normal field polarity (direction B), hence recording two geomagnetic reversals (CR1 → CN1 and CN1 → CR2). Despite the fact that the components A and C mean directions are statistically different, both directions were combined in constructing the reversal test versus the component B mean direction, which marginally passes with a "C" classification [*McFadden and Lowes*, 1981; *McFadden and McElhinny*, 1990]. Based on this result, the authors concluded that the apparent reversal asymmetry is a result of plate motion rather than the effect of a long-standing nondipole field. We note, however, that the mean direction for component B was based on only five sites, two of which have values of α_{95} greater than 15°. Eliminating these sites would leave only three normal directions, too few to perform a robust reversal test.

Based on their paleomagnetic data, the authors also proposed a model in which the eastern part of the complex (intrusive center A) was emplaced first, followed by the intrusion of the western gabbros and nepheline syenite at 1103 ± 5 Ma (intrusive center B), and finally by the emplacement of the central part (intrusive center C) at 1095 ± 5 Ma. In our paper, we will follow the same naming convention for the intrusive centers [*Lewchuk and Symons*, 1990].

3. Results

3.1. Sampling

For this study, we collected 504 core samples from 58 sites. Forty-two sites were sampled along the Trans-Canadian Highway 17, seven sites along the side roads north of town of Marathon (sites NMG1-NMG3, SL5-6, SL7, SL7-8, and SL8), four sites near town of Middleton (sites MD1-MD4), four sites north of Neys Provincial Park (sites GLG, GLG2, GLS, and GLS1), and one site taken south of the highway almost directly north from the Red Sucker Point Provincial Nature Reserve (site DM1) (Figure 2). All major lithologies of the Coldwell Complex except for granites were sampled (Table 1). Our sites SL5-SL33 and CLD1-CLD6 generally replicate the sampling localities of *Lewchuk and Symons* [1990]. However, we did not find any traces of the original sampling which could have been destroyed by widening the highway after 1990. Consequently, at many sites our samples were taken from fresher road cut outcrops.

3.2. Rock Magnetism

Samples from all sites were examined for their rock magnetic properties (Table 1). Magnetic hysteresis parameters (coercivity, H_c; coercivity of remanence, H_{cr}; saturation remanence, M_{rs}; and saturation magnetization, M_s) were measured using a Model 2900 Princeton Measurement Corporation Alternating Gradient Field Magnetometer. All measured magnetic hysteresis loops have a regular shape and indicate the presence of a low-to-intermediate coercivity magnetic phase (Figures 3a–3c). Magnetic hysteresis data did not reveal any signature of magnetically hard phases such as hematite. The M_{rs}/M_s and H_{cr}/H_c ratios suggest a pseudo-single domain (PSD) and multidomain (MD) magnetic carrier in our samples (Figure 3d) [*Day et al.*, 1977]. The relatively large scatter in the hysteresis data (Figure 3) likely reflects complex mineralogical compositions of the Coldwell Complex rocks (for example, the simultaneous presence of different iron oxides and sulfides). No systematic differences in magnetic hysteresis parameters between different lithologies were observed; however, on average, mafic rocks (gabbros, diabase, and basalt) plot closer to the single-domain region of the Day plot suggesting relatively smaller grain size of magnetic minerals.

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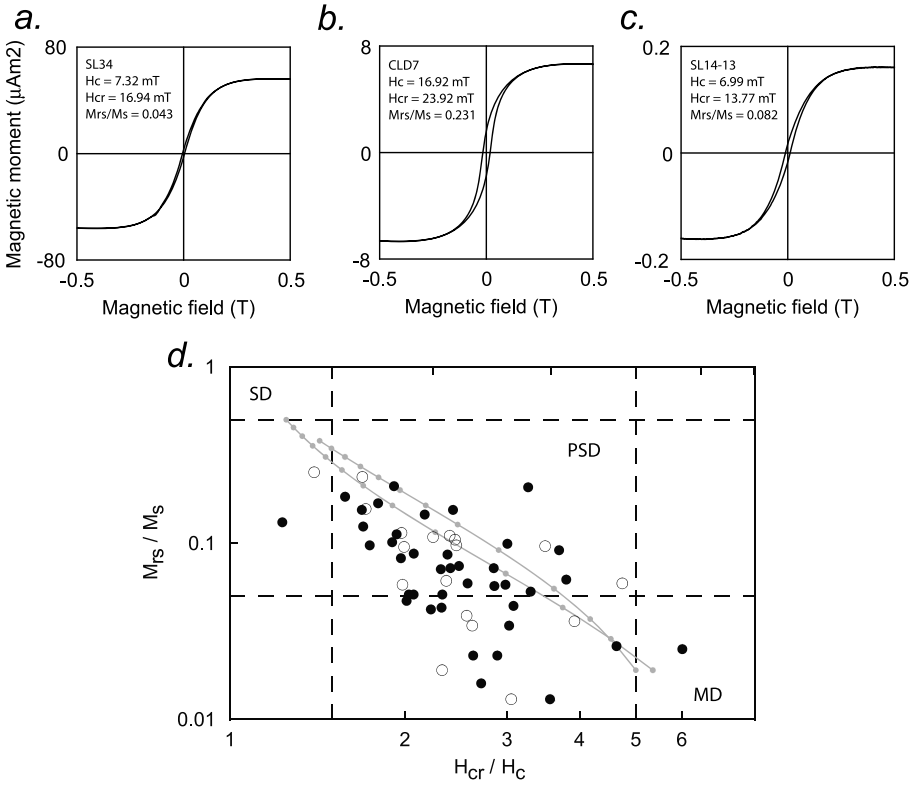


Figure 3. Magnetic hysteresis properties measured from the Coldwell Complex rocks. (a–c) Typical magnetic hysteresis loops after paramagnetic slope correction. (d) Day plot [*Day et al.*, 1977]; solid (open) symbols indicate samples from sites that yielded (failed to provide) acceptable paleomagnetic data. Abbreviations are the following: SD, single domain; PSD, pseudo-single domain; MD, multidomain; M_{rs}, saturation remanence; M_s, saturation magnetization; H_c, coercivity field; and H_{cr}, coercivity of remanence. Grey lines show SD-MD mixture models from *Dunlop* [2002].

Temperature dependences of low-field magnetic susceptibility, $\kappa(T)$, were measured upon cycling the samples from room temperature to 700°C (in argon) using an AGICO MFK-1FA magnetic susceptibility meter equipped with a high-temperature furnace and a cryostat. The $\kappa(T)$ curves were also measured upon heating from −192°C to room temperature before and after the high-temperature thermomagnetic runs. Several types of thermomagnetic $\kappa(T)$ curves were observed. Most gabbros from both eastern and western part of the complex yielded (nearly) reversible thermomagnetic curves, revealing the presence of a single magnetic phase with a Curie temperature between 570 and 585°C, suggesting magnetite or low-Ti titanomagnetite as a magnetic remanence carrier (Figures 4a and 4b). A characteristic peak observed at −153°C, associated with the Verwey transition [*Verwey*, 1939], further indicates that the magnetite is nearly stoichiometric (Figures 4a and 4b). For some gabbro samples, the $\kappa(T)$ curves also suggest the presence of a magnetic phase with Curie temperature at ~670°C, most likely hematite. These curves are slightly less reversible most likely due to reduction of hematite into a cubic ferromagnetic phase (similar to magnetite) upon heating, in argon.

Samples of ferroaugite syenite from the eastern part of the complex yielded irreversible $\kappa(T)$ curves (Figure 4d). The Verwey transition seen on the low-temperature $\kappa(T)$ curves measured before high-temperature runs together with the observed Curie temperatures near 590°C indicates magnetite as the dominant magnetic carrier in these rocks. However, many of these samples also showed a inflection on the heating $\kappa(T)$ leg at ~330°C indicating the presence of another magnetic phase which we interpreted as monoclinic pyrrhotite. The $\kappa(T)$ irreversibility for ferroaugite syenite samples is likely caused by heating-induced conversion of pyrrhotite into magnetite [e.g., *Dekkers*, 1990; *Bina and Daly*, 1994] which is supported by an ~50% increase in susceptibility at room temperature and stronger Verwey transitions observed after high-temperature runs (Figure 4d).

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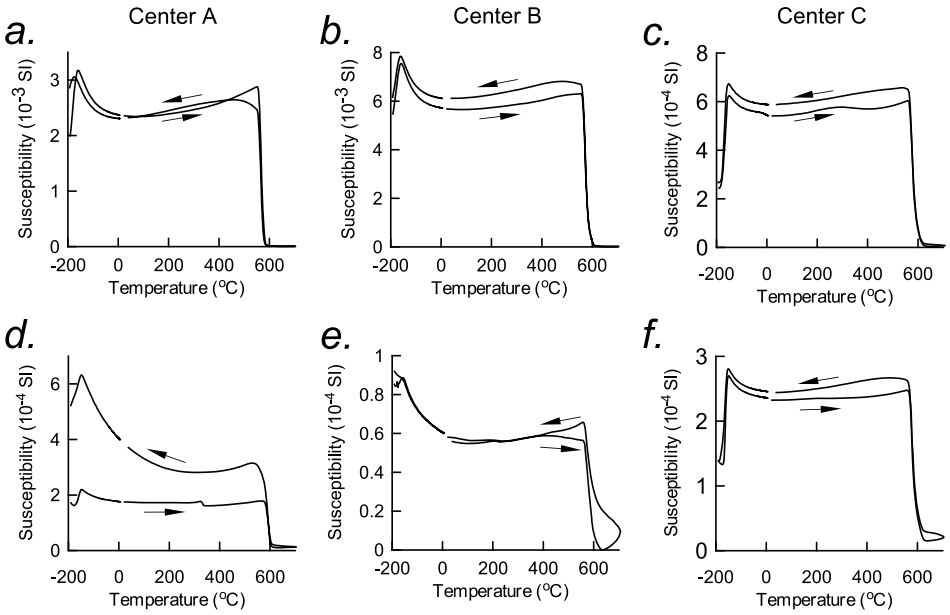


Figure 4. Typical dependence of low-field magnetic susceptibility (κ) versus temperature for Centers (a, d) A, (b, e) B, and (c, f) C. The arrows show the direction of temperature change during high-temperature runs (heating or cooling). The lithologies shown are gabbro (Figures 4a and 4b), amphibole syenite (Figure 4c), augite syenite (Figure 4d), basalt (Figure 4e), and quartz syenite (Figure 4f).

Amphibole syenites and quartz syenites from the central part of the complex yielded nearly reversible $\kappa(T)$ curves (Figures 4c and 4f) similar to those of gabbros, also suggesting magnetite as the principal magnetic carrier. However, more pronounced Verwey transitions and less pronounced Hopkinson peaks in syenite samples suggest larger multidomain magnetite [e.g., *Dunlop*, 1974] which is generally consistent with the magnetic hysteresis data. Some $\kappa(T)$ curves showed a bump at ~300–350°C on the heating $\kappa(T)$ leg which disappeared on the cooling $\kappa(T)$ leg. Such a behavior may represent temperature-induced unmixing of homogeneous titanomagnetite grains into a high-Ti and magnetite (low-Ti) phases [e.g., *Smirnov et al*., 2005] or be a result of relaxation of stress-related pinning of magnetic domain walls by heating [e.g., *Kosterov and Prévot*, 1998]. In addition, in many of these samples, a magnetic phase with Curie temperatures above 600°C was observed (e.g., Figure 4f). We interpret this phase as hematite and/or titanohematite.

The heating $\kappa(T)$ leg measured from a sample of pillow lava (CCW2, Center B) (Figure 4e) indicates the presence of two magnetic phases with Curie temperatures of ~590–600°C and ~700°C, but only the former phase is observed on the cooling leg. The low-temperature $\kappa(T)$ curve of this sample measured before the high-temperature run has an inflection point at ~ −160°C, suggesting that the ~590–600°C phase may be cation-deficient (oxidized) magnetite. The other phase can be either hematite formed by conversion of cation-deficient magnetite or maghemite, which is reduced to nearly magnetite composition by heating in argon. This interpretation is consistent with no significant change of magnetic susceptibility observed at room temperature and a more pronounced Verwey transition peak on the low-temperature $\kappa(T)$ curve measured after the high-temperature measurement (Figure 4e).

In addition to the rock magnetic investigation, we examined the opaque mineralogy of several representative samples on polished samples using a Philips XL40 environmental scanning electron microscope equipped with an energy dispersive spectra (EDS) detector. Backscattered electron imaging was used to identify oxide grains. The composition of the oxide grains was determined by means of energy dispersive spectrometry. The spectra were measured at a 15 kV accelerating voltage, which is optimal for excitation of the Fe K shell.

Most of the identified oxide minerals in both gabbros and syenites were relatively large (several hundreds of microns to $>$1 mm) grains, containing one or several subordinate sets of trellis- or sandwich-type lamellae [*Haggerty*, 1991] (Figures 5a, 5b, and 5d–5h). The EDS analyses showed that the bright areas separated by lamellae in the oxide grains represent an iron oxide phase with a very low titanium content (Figure 5j). The composition of the darker phase (lamellae) are consistent with nearly ilmenite or ulvospinel composition

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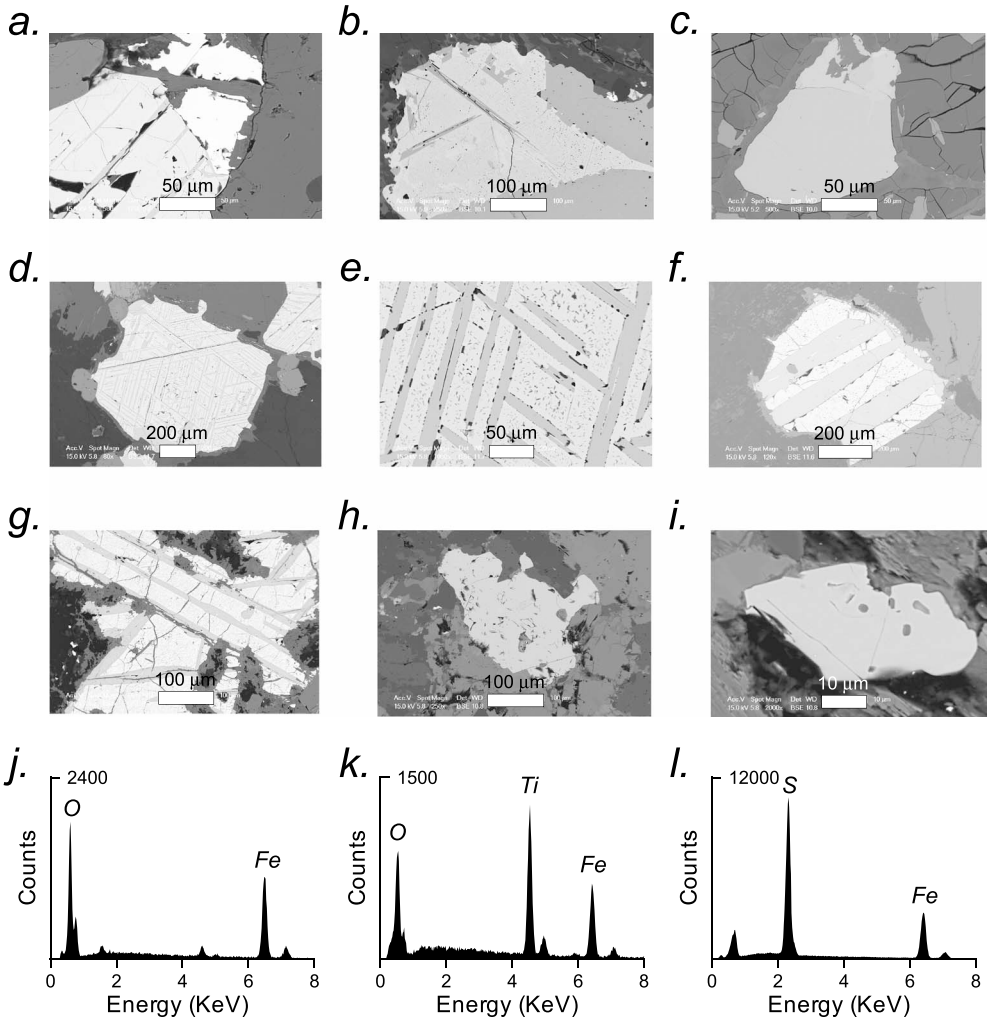


Figure 5. (a–i) Backscattered electron images of typical magnetic grains (lighter areas) of the representative Coldwell Complex samples from Centers (a–c) A, (d–f) B, and (g–i) C. Site CLD4 (gabbro), the lightest area in the center and top correspond to an iron sulfide phase (Figure 5a); Site SL15 (augite syenite) (Figures 5b and 5c); Site SL18 (gabbro) (Figures 5d and 5e); Site SL48 (nepheline syenite) (Figure 5f); Site GLG2 (gabbro) (Figure 5g); Site SL28 (amphibole syenite) (Figures 5h and 5i). Typical energy dispersive spectra from the (j) low-Ti and (k) high-Ti phases, interpreted as magnetite and ilmenite, respectively (see text). (l) Typical energy dispersive spectrum from an iron sulfide phase.

(Figure 5k). In some grains, ilmenite was partially (<5% of the total volume) replaced with sphene. We note that an exact measurement of the iron and titanium content was not possible because of the relatively large interaction volume of the electron beam (~2 μm). As a result, the EDS for both Fe-rich and Ti-rich phases may include a "contamination signal" from the surrounding and/or underlying regions of the opposite phase. In addition to the heterophase Fe-Ti oxides, some samples contained homogeneous grains (Figures 5a, 5c, and 5i). The EDS analyses indicate an iron sulfide composition of these grains (Figure 5l) consistent with the presence of pyrrhotite in ferroaugite syenites of Center A suggested by thermomagnetic analyses. Overall, rock magnetic analyses and interpretations are consistent with the results of electron microscopy.

3.3. Paleomagnetic Results

Magnetic remanence was measured with a 2G Enterprises 760-R superconducting rock magnetometer equipped with an alternating field (AF) demagnetizing unit. After measurement of a natural remanent magnetization (NRM), thermal and AF demagnetization experiments were performed on sister specimens prepared from each core sample. Thermal demagnetization was performed in an inert (nitrogen) atmosphere using an ASC TD-48SC thermal specimen demagnetizer. Progressive demagnetization (typically, 12–15 steps)

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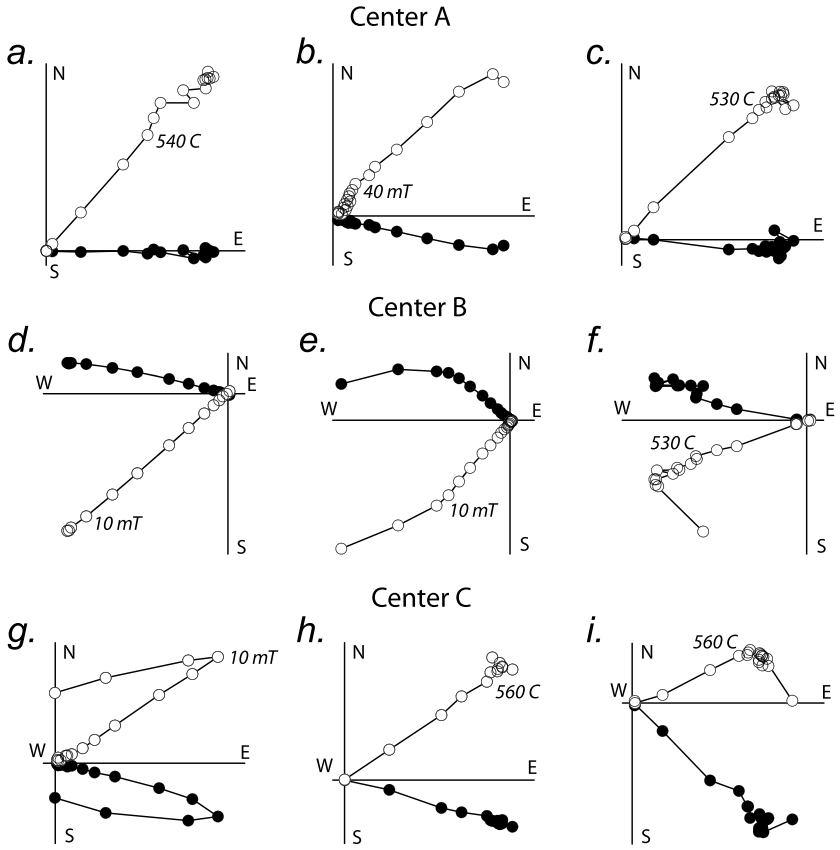


Figure 6. Examples of vector endpoint diagrams for the Coldwell Complex samples (open symbols are data points projections on vertical plane; solid symbols are projections on horizontal plane). Lithologies shown are (a, b) gabbro, (c) augite syenite, (d) basalt, (e, f) gabbro, (g) amphibole syenite, (h) quartz syenite, and (i) nepheline syenite.

was carried out until the magnetic intensity of the specimens fell below noise level or until the measured remanence directions became erratic and unstable.

Nearly all measured samples manifested a two-component NRM (Figure 6). A low-coercivity/temperature component with variable directions was removed by heating to 250–300°C or by a 10–15 mT alternating field. The characteristic remanent magnetization (ChRM) component was identified by straight linear demagnetization trajectory toward the origin on a vector endpoint diagram.

The ChRM was typically demagnetized (to less than 5% of its initial value) by 575–590°C or by 60–70 mT. The ChRM directions were calculated using principal component analysis [*Kirschvink*, 1980]. The best fit line was used if it was defined by at least five consecutive demagnetization steps that trended toward the origin and had a maximum angle of deviation less than 10°.

For the majority of samples, the ChRM directions based on AF and thermal demagnetization were statistically indistinguishable and were averaged for further calculations (Tables 2). For 15 sites, only one of the demagnetization methods provided meaningful results (Table 2). The site mean and group mean paleomagnetic directions were calculated using Fisher statistics [*Fisher*, 1953]. A site mean was accepted if the direction was obtained from three or more samples and the 95% confidence circle (α_{95}) was not greater than 15°. Directional data from 40 sites met these acceptance criteria. For six of the rejected sites, all measured specimens showed erratic demagnetization trajectories of NRM on the vector endpoint diagram. Data from additional 11 sites were rejected due to a large scatter of the ChRM directions of individual samples (i.e., $\alpha_{95} > 15°$) or an insufficient number of successful samples per site ($N < 3$) (Table 2). The remaining rejected site mean direction for site CLD9 indicated remagnetization by the present-day geomagnetic field (Table 2).

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Table 2. Summary of Paleomagnetic Results From the Coldwell Complex[a]

Site ID	λ_s, °N	Φ_s, °E	N(n)	B(b)	Method	D (°)	I (°)	α_{95} (°)	k	λ_p, °N	Φ_p, °E
					Center A						
CLD1	48.713	−86.322	3(7)		AF + T	116.0	−61.5	7.4	187	46.2	200.6
CLD3	48.719	−86.673	8(9)		AF + T	125.7	−73.8	8.6	37	57.5	224.3
CLD4	48.727	−86.328	14(17)		AF + T	103.7	−66.4	4.4	76	42.0	214.4
CLD5	48.729	−86.329	7(11)		AF + T	116.9	−73.3	4.4	76	52.9	224.1
CLD6	48.742	−86.334	9(10)		AF + T	106.3	−67.2	8.5	34	44.0	214.5
NMG1	48.847	−86.484	9(10)		AF + T	99.1	−51.9	12.3	32	29.5	200.4
SL10	48.768	−86.374	5(6)		AF + T	118.5	−70.7	6.4	114	52.8	217.2
SL14	48.780	−86.420	6(11)		AF + T	125.5	−49.2	11.0	32	45.1	179.7
SL15	48.773	−86.393	5(12)		T	118.4	−68.5	11.0	43	51.7	212.2
SL33	48.785	−86.428	8(9)		AF + T	118.6	−66.7	3.9	174	50.9	208.3
SL34	48.726	−86.327	4(7)		AF + T	138.5	−75.6	9.1	73	63.4	231.1
SL5-6	48.732	−86.330	8(8)		AF + T	98.0	−63.5	6.7	61	36.7	213.0
SL7	48.756	−86.310	5(6)		AF + T	113.7	−68.8	6.6	110	49.2	214.6
SL7-8	48.755	−86.311	4(6)		AF	127.4	−62.4	11.0	52	54.0	196.0
	Mean			**14(19)**		**115.1**	**−66.1**	**4.4**	**82**	**48.9**	**209.8**
										A_{95} = 6.0	**K = 45**
CLD2[b]	48.708	−86.303	0 (13)		AF + T	-	-	-	-		
CLD9[b]	48.768	−86.373	3(12)		AF + T	351.6	63.4	13.6	55		
SL8[b]	48.754	−86.310	0(7)		AF + T	-	-	-	-		
NMG2[b]	48.800	−86.314	8(8)		AF + T	129.4	−48.1	35.2	2.7		
NMG3[b]	48.787	−86.316	7(8)		AF + T	171.6	−44.0	58.4	1.7		
					Center B						
CCW1	48.814	−86.673	5(11)		T	267.0	53.2	14.3	23	22.9	209.0
CCW2	48.817	−86.684	8(11)		AF + T	293.3	51.0	5.9	77	38.1	190.1
CLD7	48.770	−86.549	4(12)		AF	318.0	57.3	9.3	73	58.1	179.6
SL13E[c]	48.796	−86.453	8(8)		T	303.5	62.6	4.8	120	51.6	198.3
SL18	48.799	−86.651	6(8)		AF + T	299.2	49.8	6.0	107	41.3	184.8
SL19	48.796	−86.646	5(7)		AF + T	295.5	58.5	10.0	46	44.1	196.7
SL20	48.800	−86.636	6(8)		AF + T	308.2	59.3	14.0	21	52.8	189.9
SL37	48.802	−86.626	6(8)		T	321.3	54.4	4.9	154	58.5	171.9
SL47	48.817	−86.682	7(8)		AF	287.7	58.7	10.8	28	39.2	201.5
SL49	48.797	−86.651	6(8)		AF	289.6	55.1	5.2	142	38.2	196.5
	Mean w/o SL13E			**9(15)**		**297.5**	**56.3**	**6.4**	**66**		
	Mean			**10(16)**		**298.0**	**56.9**	**5.8**	**70**	**44.9**	**193.2**
										A_{95} = 8.0	**K = 37**
MD1[b]	48.798	−86.654	8(8)		AF + T	3.5	0.2	62.8	1.5		
MD2[b]	48.797	−86.656	12(15)		AF + T	313.3	8.4	43.3	1.6		
MD3[b]	48.795	−86.658	9(12)		AF + T	54.2	−43.5	53.2	1.7		
MD4[b]	48.792	−86.661	3(12)		AF + T	349.0	12.2	84.5	1.7		
SL26[b]	48.807	−86.662	0(12)		AF + T	-	-	-	-		
SL48[b]	48.798	−86.640	4(8)		AF + T	316.3	27.9	27.7	9		
					Center C						
CLD11	48.784	−86.596	4(7)		AF	129.4	−49.2	7.7	107	47.9	176.7
GLS	48.824	−86.485	5(5)		AF + T	92.1	−57.2	14.9	23	28.7	209.3
GLG	48.821	−86.485	4(6)		AF + T	131.5	−43.8	8.4	90	46.0	229.8
GLS2	48.824	−86.485	3(4)		AF	110.2	−49.6	7.9	162	35.3	191.2
CLD8	48.765	−86.525	3(6)		AF	131.2	−64.5	6.3	258.7	44.0	214.5
SL11	48.792	−86.487	5(10)		AF + T	77.1	−68.9	4.8	205	30.4	229.9
SL13	48.793	−86.463	6(6)		AF + T	120.4	−60.5	4.4	191	48.4	196.7
SL14-13	48.779	−86.420	8(9)		AF + T	89.8	−69.1	3.1	284	36.6	224.5
SL21	48.789	−86.611	6(8)		AF + T	120.7	−57.5	7.0	77	46.8	192.2
SL25	48.793	−86.501	4(6)		AF	110.6	−57.0	1.7	2315	40.0	198.0
SL28	48.802	−86.626	6(12)		AF + T	118.6	−45.4	7.7	64	38.4	181.7
SL32	48.795	−86.495	8(8)		AF + T	99.4	−75.2	6.4	67	45.7	232.2
SL41	48.797	−86.445	6(8)		AF	132.2	−67.9	8.3	56	59.7	206.0
SL44	48.764	−86.526	3(9)		AF + T	114.0	−59.7	7.5	56	43.8	199.3
SL45	48.772	−86.513	8(10)		AF	108.5	−69.7	9.1	38	46.8	218.0
SL46	48.795	−86.497	3(9)		AF	114.1	−76.3	5.9	292	52.6	232.3

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Table 2. (continued)

Site ID	λ_s, °N	Φ_s, °E	*N*(n)	*B*(b)	Method	*D* (°)	*I* (°)	α_{95} (°)	*k*	λ_p, °N	Φ_p, °E
Mean				**16(23)**		**114.6**	**−61.6**	**5.8**	**42**	**45.6**	**203.7**
										A_{95} **= 7.6**	***K*** **= 24**
GLG2[b]	48.824	−86.485	0(6)		AF + T	-	-	-	-		
CLD10[b]	48.780	−86.581	6(8)		AF + T	262.8	−66.7	58.8	1.8		
DM1[b]	48.789	−86.480	13(13)		AF + T	208.3	−28.1	51.3	1.5		
SL23[b]	48.772	−86.556	2(8)		AF + T	-	-	-	-		
SL30[b]	48.794	−86.489	0(7)		AF + T	-	-	-	-		
SL31[b]	48.792	−86.491	2(8)		AF + T	-	-	-	-		
SL40[b]	48.792	−86.468	0(8)		AF + T	-	-	-	-		
					Centers A and C Combined						
Mean				**30(42)**		**114.8**	**−63.7**	**3.6**	**54**	**47.2**	**206.5**
										A_{95} **= 4.8**	***K*** **= 31**

[a]The site latitude and longitude are λ_s and Φ_s, *N* is the number of samples used to calculate the site mean declination (*D*) and inclination (*I*); *n* is the number of measured samples per site; α_{95} and *k* are the 95% confidence circle and the concentration parameter for paleomagnetic directions [*Fisher*, 1953]. *B* is the number of sites used for a between-site group mean direction; *b* is the total number of sites sampled per magmatic episode. The column "Method" indicates the demagnetization method used to obtain the site mean directions: alternating field (AF), thermal (T), or combined (AF + T) (see text). The latitude and longitude of the virtual geomagnetic pole (VGP) are λ_p and Φ_p. A_{95} and *K* are the 95% confidence circle and the concentration parameter for VGP distribution.
[b]Paleomagnetic data not included into paleomagnetic analysis.
[c]This site is located within the central part of the Coldwell Complex (see text).

The accepted site mean directions were grouped according to the classification of intrusive centers used by *Lewchuk and Symons* [1990]. Fourteen (out of 19 sampled) sites collected from rocks of Center A (eastern gabbro and ferrosyenite) yielded well-defined reversed ChRM directions with a group mean direction of $D = 115.1°$, $I = -66.1°$ ($\alpha_{95} = 4.4°$, $k = 82$) (Figure 7a and Table 2). The corresponding mean virtual geomagnetic pole (VGP) is located at $P_{lat} = 48.9°$N, $P_{long} = 209.8°$E ($A_{95} = 6.0°$, $K = 45$) (Pole CCA, Figure 8a and Table 2).

Nine of 15 sampled sites from Center B yielded well-defined normal polarity site mean directions with a group mean of $D = 297.5°$, $I = 56.3°$ ($\alpha_{95} = 6.4°$, $k = 66$) (Table 2). In addition, site SL13E representing a thin mafic dyke cutting through syenite in the central part of the complex resulted in a normal polarity ChRM direction ($D = 303.5°$, $I = 62.6°$, $\alpha_{95} = 4.8°$). Accordingly, the SL13E direction has been included in calculation of the group mean direction for Center B ($D = 298.0°$, $I = 56.9°$, $\alpha_{95} = 5.8°$, $k = 70$, $N = 10$) (Figure 7b and Table 2). The corresponding mean VGP is located at $P_{lat} = 44.9°$N, $P_{long} = 193.2°$E ($A_{95} = 8.0°$, $K = 37$) (Pole CCn, Figure 8b and Table 2).

We note that while the accepted site mean directions for Center B pass the test for Fisher distribution [*Fisher et al*., 1987], one of the sites (CCW1) has a more westerly declination and plots away from the rest of site mean directions (Figure 7b). This site has the largest $\alpha_{95} = 14.3°$. If this direction is excluded, the new group mean does not significantly change ($D = 301.7°$, $I = 56.8°$, $\alpha_{95} = 5.0°$, $k = 108$, $N = 9$) with the pole located at $P_{lat} = 47.2°$N, $P_{long} = 190.9°$E ($A_{95} = 6.7°$, $K = 60$). Our preferred explanation is that the deviation of the CCW1 direction from the rest of the group is due to paleosecular variation. Therefore, in our further analysis, we use the group mean direction calculated from all 10 sites.

Sixteen site mean directions (of 23 sampled sites) of Center C yielded reversed polarity ChRM directions with a group mean direction of $D = 114.6°$, $I = -61.6°$ ($\alpha_{95} = 5.8°$, $k = 42$) (Figure 7c) and the corresponding mean VGP position located at $P_{lat} = 45.6°$N, $P_{long} = 203.7°$E ($A_{95} = 7.6°$, $K = 24$) (Pole CCC, Figure 8a and Table 2). The group mean directions from Centers A and C are statistically indistinguishable (the group mean direction from Center C plots within the other direction α_{95} [e.g., *Fisher et al*., 1987]) and were combined resulting in a group mean direction of $D = 114.8°$, $I = -63.7°$ ($\alpha_{95} = 3.6°$, $k = 54$, $N = 30$) (Figure 7d and Table 2). The corresponding mean VGP position is located at $P_{lat} = 47.2°$N, $P_{long} = 206.5°$E ($A_{95} = 4.8°$, $K = 31$) (Pole CCr, Figure 8 and Table 2).

In order to evaluate the symmetry of normal and reversed group mean directions recorded by rocks of the Coldwell Complex, we used the reversal test of *McFadden and McElhinny* [1990]. In this test, information

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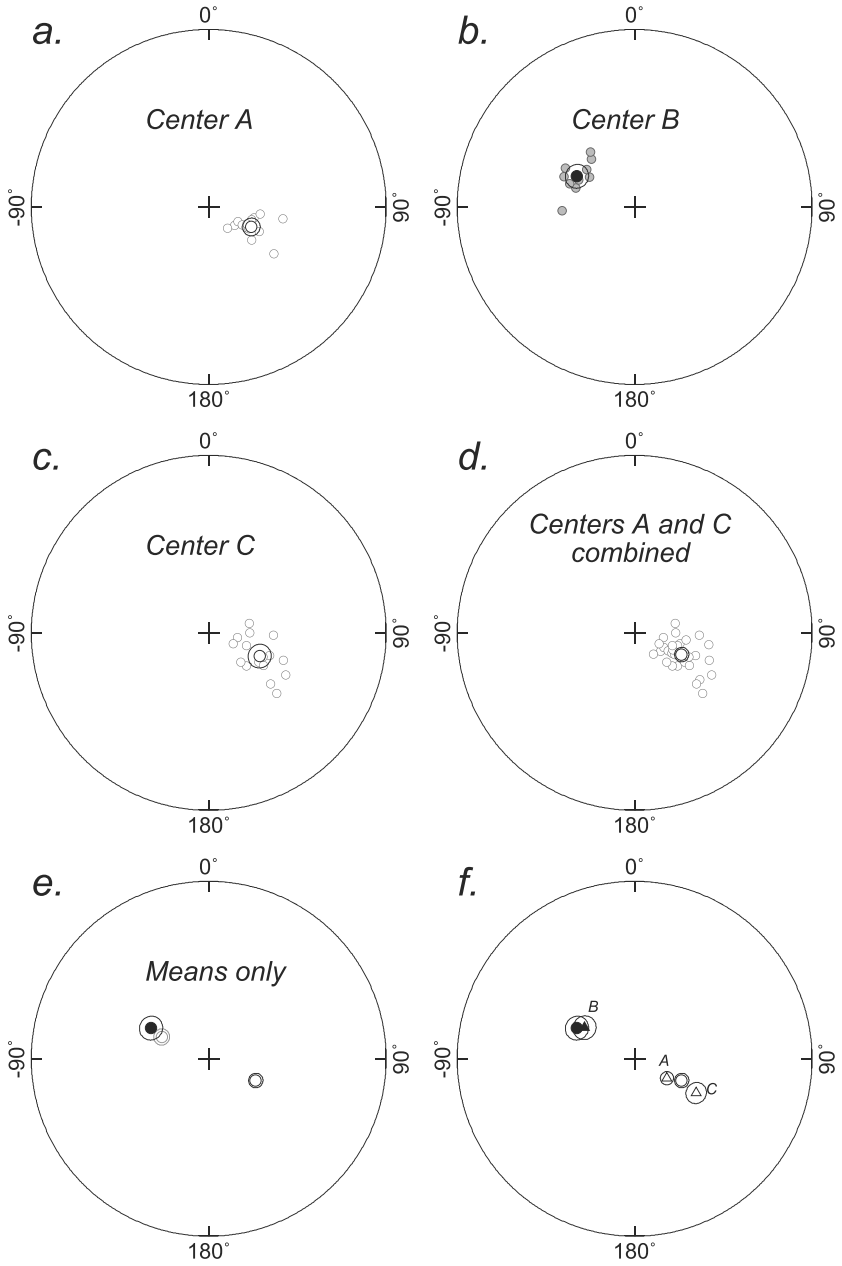


Figure 7. (a–c) Paleomagnetic results from Centers A, B, and C of the Coldwell Compex, respectively. Equal area plots show the accepted paleomagnetic site mean directions (grey circles) and their group mean direction with the 95% confidence circle (α_{95}). Filled and open circles correspond to normal and reversed polarity, respectively. (d) Combined site mean directions (grey circles) for Centers A and C and their group mean (open circle) with the 95% confidence circle. (e) The group means for normal (Center B, solid circle) and reversed (Centers A and C, open circle) polarity direction obtained in this study. The grey circle shows the direction exactly symmetrical to the reversed direction. (f) Comparison of the results of this study with the results by *Lewchuk and Symons* [1990]. Open and solid triangles show their group mean directions for Centers A and C and for Center B, respectively. The open and solid circles are the same as in Figure 7e.

about the angular dispersion of the two distributions being compared is used to calculate the critical angle γ_c between the means of each set of observations at which the null hypothesis that the data share a common mean would be rejected at 95% confidence. If the angle between the two means is less than the critical angle, the test is positive. *McFadden and McElhinny* [1990] proposed a simple classification of a positive reversal test: classification A if $\gamma_c \leq 5°$, B if $5° < \gamma_c \leq 10°$, C if $10° < \gamma_c \leq 20°$, and "Indeterminate" if $\gamma_c > 20°$.

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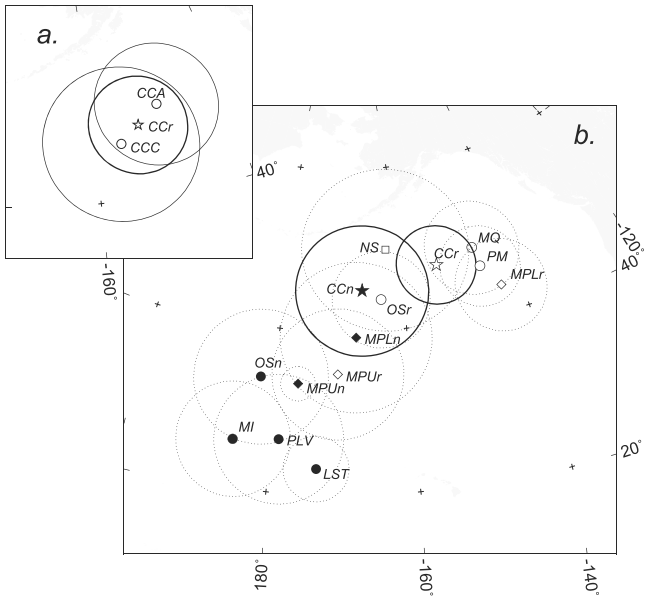


Figure 8. (a) Group mean virtual geomagnetic poles for Centers A (CCA) and C (CCC) of the Coldwell Complex (open circles) and their 95% confidence circles (A_{95}). Star shows the paleomagnetic pole (CCr) calculated from combined site mean directions of Centers A and C (see text). (b) Selected paleomagnetic and mean VGP poles from the Midcontinent Rift sequences (see text). The open/solid symbols correspond to the reversed/normal polarity of paleomagnetic directions. The open and closed stars show the CCr pole and the mean VGP pole (CCn) from the normally magnetized rocks of Center B (this study). Diamond shows the mean VGP pole calculated for the lower reversed section (MPlr), the lower normal section (MPLn), the upper reversed section (MPUr), and the upper normal (MPUn) section of the Mamainse Point lava flow sequence [*Swanson-Hysell et al*., 2009]. The other poles are from the Powder Mill basalts (PM) [*Palmer and Halls*, 1986], the Lower and Upper Osler Volcanics (OSr and OSn) [*Halls*, 1974], the Lower North Shore Volcanics (NS) [*Halls and Pesonen*, 1982], the Marquette dike swarm (MQ) [*Pesonen and Halls*, 1979], the Portage Lake Volcanics (PLV) [*Halls and Pesonen*, 1982], the Lake Shore Traps (LST) [*Kulakov et al*., 2013], and the Michipicoten Island flows (MI) [*Palmer and Davis*, 1987].

The reversed site mean directions calculated for Centers A and C pass the reversal test with respect to the normal site mean direction of Center B with classification B ($\gamma_c = 9.3°$) and A ($\gamma_c = 5.0°$), respectively. The combined direction from Centers A and C passes the reversal test against the Center B direction with classification B ($\gamma_c = 7.0°$).

4. Discussion

4.1. Do the Coldwell Complex Rocks Faithfully Record the Earth's Magnetic Field Direction?

The results of our rock magnetic analyses do not reveal any obvious correlation of the magnetic characteristics of our samples with their ability to preserve primary paleomagnetic directions or with the quality of paleomagnetic data (Figures 3 and 4 and Table 1). However, we note that eight of 18 sites that failed to yield acceptable paleomagnetic data were collected from natural outcrops affected by weathering, while all the successful sites were sampled from fresh road cut exposures.

The primary origin of characteristic remanent magnetization of the accepted sites is supported by the positive reversal tests between the normal and reversed directions and by the statistical similarity of ChRM directions recorded by different lithologies within each intrusive center. Furthermore, the positive reversal tests suggest that the group mean paleomagnetic directions for each center are not significantly biased by insufficient sampling of the paleosecular variation.

To further evaluate whether our paleomagnetic data represent the time-averaged geomagnetic field, we calculated the angular dispersion (S) of virtual geomagnetic poles (VGP) for each center:

$$S^2 = \frac{1}{N-1}\sum_{i=1}^{N}\Delta_i^2,$$

where N is the number of individual VGPs and Δ_i is the angle between the ith VGP and the group mean pole. The dispersion values were corrected for within-site dispersion (S_w) related to intrinsic variation and experimental uncertainty using:

$$S_b^2 = S^2 - S_w^2/n,$$

where S_b is the true (between-site) dispersion and n is the average number of samples per site [*Doell*, 1970].

The confidence interval of S_b were estimated using the N-1 jackknife method [*Efron*, 1982]. The calculated values, $S_{b_A} = 13.2 \pm 5.2°$ ($N = 14$), $S_{b_B} = 12.1 \pm 5.5°$ ($N = 10$), and $S_{b_C} = 15.9 \pm 3.6°$ ($N = 16$) are in a good

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Table 3. Comparison of Paleomagnetic Results of This Study With the Results by *Lewchuk and Symons* [1990][a]

		N	*D* (°)	*I* (°)	*k*	α_{95} (°)	Φ_p (°E)	λ_p, (°N)	A_{95} (°) or δ_p/δ_m (°)
Center A	This study	14	115.1	−66.1	89	4.4	209.8	48.9	6.0
	LS90 only	11	117.8	−71.1	132	4.0	218.2	52.6	6.1/7.0
	Robertson	8	125.3	−70.6	7.6	6.4	215.1	56.2	9.6/11.1
	LS90 + R	19	121.0	−70.9	104	3.3	217.0	54.2	5.0/5.7
Center B	This study	10	298.0	56.9	70	5.8	193.2	44.9	8.0
	LS90 only	5	313.3	65.3	47.3	11.3	198.5	59.6	14.8/18.3
	Robertson	6	264.5	55.2	193.5	5.9	41.4	6.0	8.4
	LS90 + R	11	301.8	60.1	49	5.8	195.1	49.1	7.5/10.0
Center C	This study	16	114.5	−61.6	48	5.8	203.7	45.6	7.6
	LS90 only	9	119.1	−54.3	101	5.1	189.6	43.3	5.0/7.1
	Robertson	6	114.6	−70.5	91	7.1	217.7	50.6	11.0
	LS90 + R	15	117.9	−60.7	50.3	5.4	202.7[b]	45.3[b]	45[b]

[a] *N*, number of sites; *D* and *I* are magnetic declination and inclination, respectively; α_{95} and *k* are the 95% confidence circle and the concentration parameter for paleomagnetic directions [*Fisher*, 1953]; λ_p and Φ_p are the latitude and longitude of the mean virtual geomagnetic pole (VGP). A_{95} and *K* are the 95% confidence circle and the concentration parameter for VGP distribution. LS90, *Lewchuk and Symons* [1990 data]; R, W. A. Robertson's unpublished data (cited in *Lewchuk and Symons* [1990]).
[b] Due to absence of the precise coordinates of sampling sites in the prior studies, the pole was calculated as a mean of LS90 and R mean paleopoles.

agreement with the VGP angular dispersion (9.0°–14.6°) reported from other Keweenawan rock sequences [e.g., *Halls and Pesonen*, 1982] and are statistically similar to the mean value of $S \approx 14.0°$ (for paleolatitude of 45°) calculated for the 1.0–2.2 Ga interval [*Smirnov et al*., 2011]. We note that if site CCW1 is excluded, then the angular dispersion for Center B is lowered: $S_{b_B} = 9.1 \pm 4.8°$ ($N = 9$). Given the relatively small number of site mean directions ($N = 10$), the data set from Center B may not entirely represent the time-averaged field.

We note that some of the site mean directions used to calculate *S* may represent the same vector of geomagnetic field (if the corresponding sites cooled through their magnetization blocking temperatures within a few years from each other). Although such a scenario is unlikely taking into account relatively slow cooling of the complex, if true, our estimates would represent the lower limit of *S*. On the other hand, there is no evidence of extensive faulting and/or significant relative block rotations which could have resulted in an artificial increase of *S*. Overall, we feel that the general similarity of our angular dispersion estimates to those obtained from other MCR rocks indicates that our paleomagnetic data adequately represent the paleosecular variation of geomagnetic field.

As for most large-scale intrusions, it is difficult to gauge the structural attitude of the Coldwell Complex rock units which naturally raises the question whether our paleomagnetic results could have been affected by potential large-scale relative movements between the different rock units or rotation of the entire complex. However, significant interblock movements can be ruled out by the positive reversal tests as well as by the statistical similarity of the paleomagnetic directions obtained from different lithological units. The possibility of a significant tilting of the entire complex can also be ruled out as the basaltic lava flows of the Giordie-Coubran Lake are basically flat lying [*Kulakov et al*., 2012]. Large-scale tilt or rotation is also inconsistent with the observed similarity of the Coldwell Complex paleomagnetic directions with the directions from other MCR rock units (see section 4.3). Overall, based on the observations described above, we feel confident that the obtained ChRM directions faithfully represent the Earth's magnetic field that existed during the initial cooling of the complex.

4.2. Comparison With the Results by *Lewchuk and Symons* [1990]

The group mean direction for Center B reported in this paper is similar to the direction reported by *Lewchuk and Symons* [1990] (Figure 7f and Table 3). In contrast, the group mean directions for Centers A and C obtained in the prior study have significantly steeper and shallower inclinations, respectively, when compared to our group mean directions for these centers or to their combined direction (Figure 7f and Table 3).

This difference, we suggest, is probably the direct result of the laboratory techniques used by the two studies to determine a sample's ChRM. Specifically, *Lewchuk and Symons* [1990] chose two pilot samples from each site in

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their study to be completely demagnetized using either AF or thermal methods, and from that data they chose three to four demagnetization steps to demagnetize the rest of the samples from that particular site. A similar demagnetization strategy was used by W. A. Robertson whose unpublished data for the Coldwell Complex was included in the Lewchuk and Symons' paper [*Lewchuk and Symons*, 1990, Table 2]. This methodology was a perfectly acceptable technique at the time when the authors submitted their manuscript for publication. However, our experience with the rocks from the Coldwell Complex (based in part on Lewchuk and Symons' paper) is that each sample from a given site must be completely demagnetized to adequately define the sample's ChRM before calculating a site mean. This is the experimental approach most paleomagnetic laboratories are currently using. Therefore, we propose that the "semiblanket" demagnetization approach by *Lewchuk and Symons* [1990] based on pilot sample behavior may have resulted in an unremoved normal polarity secondary overprints. If correct, this would cause reversely magnetized rocks to have steeper negative inclinations and normally magnetized rock to shallower positive inclinations in a fashion first proposed by *Palmer* [1970] for the cause of asymmetrical reversals observed in Keweenawan rocks. Inspection of Table 3 supports the above hypothesis as it shows that our means for the reversely magnetized Centers A and C rocks are shallower than the combined means for the same centers published by *Lewchuk and Symons* [1990]. Additionally, partial removal of a secondary overprint may also explain the large α_{95} observed for some sites in the prior study. Furthermore, *Lewchuk and Symons* [1990] use ChRMs calculated from all specimens cut from an oriented core or hand sample as independently oriented samples. By doing so, they artificially reduce the size of the 95% confidence circle (α_{95}) of the site. In addition, counting all the specimen directions independently may have increased the statistical weight to nonrepresentative directions, thus biasing the calculated group mean directions.

We further note that *Lewchuk and Symons* [1990] reported a normal polarity direction B ($D = 300°$, $I = 57°$) from their site ("13") designated as quartz syenite in the central part of the complex, while we obtained an almost antipodal direction ($D = 120.4°$, $I = -60.5°$) from the same location (our site SL13). The reason for this discrepancy is unknown, but it is worth noting that both sites (13 and SL13) are in a close proximity to our site SL13E (a mafic dike that cuts through the syenite) that yielded a normal polarity direction ($D = 303.5°$, $I = 62.6°$). So the Lewchuk and Symons' site 13 may be equivalent to our site SL13E and if so should have been included to the group mean calculation for component B. However, in the prior study, this direction was inverted to the reversed polarity and included in the population of reversely magnetized site means with direction C.

4.3. Implications for the Coldwell Complex Evolution

Lewchuk and Symons [1990] used the asymmetry between their directions A and C to propose a model in which the eastern part of the complex (Center A) was emplaced first during a reversed polarity period, followed by emplacement of the western part (Center B) during a normal polarity period, and finally by emplacement of the central part (Center C) during the next (younger) reversed polarity period. We note that this model differs from the prior models based on geological data that suggested a monotonic westward age progression of the intrusive centers [e.g., *Mitchell and Platt*, 1977 *Mitchell and Platt*, 1978; *Currie*, 1980; *Mitchell et al.*, 1993; *Walker et al.*, 1993]. In particular, their model is inconsistent with the observation that the western rock units appear to be intrusive into most other phases of the complex [*Mitchell and Platt*, 1978].

The statistical similarity of the group mean directions for Centers A and C established in our study does not support the *Lewchuk and Symons* [1990] model but instead indicates that both centers were emplaced in a single or several magmatic pulses within the same period of a reversed geomagnetic polarity around ~1108 Ma. This interpretation is supported by the proximity of the paleomagnetic pole calculated from the combined reversed mean direction to the pole positions calculated from other reversely magnetized volcanic sequences representing the early stage of MCR development including the 1107.3 ± 1.6 Ma Powder Mill basalts [*Palmer and Halls*, 1986], the 1107 + 4/−2 Ma Lower Osler Volcanics [*Halls*, 1974], and the 1107.9 ± 1.8 Ma Lower North Shore Volcanics [*Books*, 1972; *Palmer*, 1970; *Halls and Pesonen*, 1982] (Figure 8). The concordance of our paleopole to the poles for rocks of similar age found elsewhere around Lake Superior provides an additional support that the Coldwell Complex did not experience significant tilt or rotation after its emplacement. We note that a paleomagnetic pole calculated for the 1108 ± 1 Ma intrusive center A ($P_{lat} = 54.2°$N, $P_{long} = 217.0°$E, $A_{95} = 5.4°$) by *Lewchuk and Symons* [1990] plots farther away from the group of the early MCR poles (Table 4).

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Table 4. Summary of Paleomagnetic and Geochronology Data for the Selected Rock Units Associated With the North American Midcontinent Rift, Shown in Figure 8[a]

Rock Unit (Polarity)	P_{lat} (°N)	P_{long} (°E)	A_{95} (°)	N	Paleomagnetic Study	Age (Ma)	Geochronology Study
Coldwell Complex (*R*) Center A	48.9	209.8	6.0	14	This study	1108 ± 1	*Heaman and Machado* [1992]
Coldwell Complex (*N*) Center B	44.9	193.2	8.0	10	This study	n/a	
Coldwell Complex (*R*) Center C	45.6	203.7	7.6	16	This study	n/a	
Coldwell Complex (*R*) Centers A and C	47.2	206.7	4.8	30	This study	1108 ± 1	*Heaman and Machado* [1992]
Coldwell Complex (*R*) Centers A, B, and C	46.8	203.1	4.2	40	This study	n/a	
Mamainse Point (*R*) Lower reversed 1	47.5	226.7	8.0	14	*Swanson-Hysell et al.* [2009]	n/a	
Mamainse Point (*R*) Lower reversed 2	37.5	206.7	5.1	12	*Swanson-Hysell et al.* [2009]	n/a	
Mamainse Point (*N*) Lower normal	38.0	190.8	9.7	11	*Swanson-Hysell et al.* [2009]	n/a	
Mamainse Point (*R*) Upper reversed 1	34.7	189.2	8.0	11	*Swanson-Hysell et al.* [2009]	n/a	
Mamainse Point (*N*) Upper normal	33.8	192.0	2.9	21	*Swanson-Hysell et al.* [2009]	n/a	
Powder Mill Group (*R*) (Siemens Creek Formation)	45.8	214.0	9.2	10	*Palmer and Halls* [1986]	1107.3 ± 1.6	*Davis and Green* [1997]
Lower North Shore Volcanics (*R*)	49.8	197.8	10.4	3	*Halls and Pesonen* [1982]	1107.9 ± 1.8	*Davis and Green* [1997]
North Shore Volcanics (*N*)	35.8	181.7	3.0	45	*Tauxe and Kodama* [2009]	1098.4 ± 1.9, 1099.3 ± 0.3	*Davis and Green* [1997] and *Paces and Miller* [1993]
Osler Volcanics (*R*)	43.1	194.5	5.9	25	*Halls* [1974]	1107.5 +4/−2, 1105.3 ± 2.1	*Davis and Green* [1997] and *Davis and Sutcliffe* [1985]
Osler Volcanics (*N*)	33.9	177.9	8.1	5	*Halls* [1974]	n/a	
Portage Lake Volcanics (*N*)	26.5	181.2	7.9	4	*Halls and Pesonen* [1982]	1094 ± 1.5, 1096 ± 1.8	*Davis and Paces* [1990]
Lake Shore Traps (*N*)	23.1	186.4	4.0	31	*Kulakov et al.* [2013]	1087 ± 1.6	*Davis and Paces* [1990]

[a] P_{lat} and P_{long} are the latitude and longitude of the paleomagnetic pole, A_{95} is the 95% confidence circle, and N is the number of sites; n/a = not applicable.

Our new paleomagnetic pole (CCr) merits almost perfect six points on the seven-point paleomagnetic reliability classification scale [*Van der Voo*, 1990]. The CCr pole also plots close to the mean paleomagnetic pole determined from the lower reversed Mamainse Point section (Figure 8) providing a time constraint on its age (~1108 Ma). Despite the positive reversal test, we note that the 95% confidence circle of the combined (A + C) group mean direction (inverted to the normal polarity) does not include the group mean direction for Center B, and vice versa (Figure 7e). The difference between the directions is entirely in inclination and may reflect the equatorward motion of the North American Plate. However, as discussed above, Center B direction is only based on 10 site mean directions and may not entirely average out the secular variation. For this reason we prefer not to calculate the combined pole based on all 40 normal and reversed accepted paleomagnetic directions.

The mean pole calculated from Center B sites ($D = 298.0°$, $I = 56.9°$, $\alpha_{95} = 5.8°$, $k = 70$, $N = 10$) plots southwest from the group of "reversed" poles on the North American apparent polar wander path (pole CCn, Figure 8) and is statistically similar to the mean VGP pole calculated for the lower normally magnetized section of the Mamainse Point sequence (pole MPLn, Figure 8). At the same time, our CCn pole plots north from the group of normal polarity poles that includes the ~1095 Ma Osler volcanics, the ~1096 Ma Portage Lake Volcanics, and the upper normally magnetized section of the Mamainse Point sequence (MPUn). This group of poles is "separated" from the older CCn and MPLn poles by the pole from the upper reversed polarity interval of the Mamainse Point volcanics (MPUr) [*Swanson-Hysell et al.*, 2009] and recently dated at ~1100.3 ± 0.4 Ma [*Swanson-Hysell et al.*, 2014] (Figure 8). Furthermore, normally magnetized rocks of the Mellen Complex [*Books*, 1972] have been radiometrically dated at 1102.0 ± 2.8 Ma [*Zartman et al.*, 1997]. We therefore conclude that the rocks of Center B were emplaced during the same period of normal polarity as the lower normally magnetized section of the Mamainse Point sequence (Figure 9). Therefore, the rocks of Coldwell Complex record a single reversal from a reversed to normal polarity which is equivalent to the

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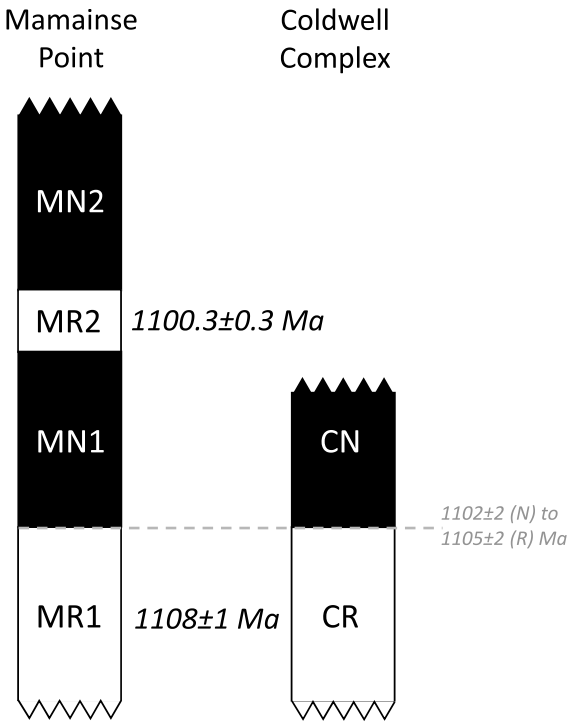


Figure 9. Simplified stratigraphic columns comparing geomagnetic reversals recorded by rocks of the Coldwell Complex and Mamainse Point (see text).

lowermost R → N polarity reversal recorded by the Mamainse Point lavas. Assuming that the next N → R reversal occurred before 1100 Ma as suggested by the age of MPUr pole, our paleomagnetic data suggest that the Coldwell Complex was emplaced within an 5–8 Ma period, which is longer than an estimate by *Heaman and Machado* [1992] but shorter than 15–20 Ma proposed by *Lewchuk and Symons* [1990].

Unfortunately, out attempts to obtain radiometric ages from the normally magnetized rocks of the Coldwell Complex were unsuccessful (K. Chamberlain, personal communication via e-mail, 2014). There is also no radiometric age available for the lower normal Mamainse Point section. The minimum age of 1105.3 ± 2.1 Ma was assigned to reversely magnetized lavas of Osler Volcanics based on the reversely magnetized Agate Point Rhyolite [*Davis and Green*, 1997]. Based on available radiometric dates, we conclude that the rocks of Center B were emplaced between ~1105 and ~1100 Ma which is in a good agreement with the period of normal polarity of geomagnetic field between 1105 ± 2 and 1102 ± 2 Ma suggested by *Davis and Green* [1997].

Our paleomagnetic data further confirm the model of westward migration of the magma source during the evolution of the Coldwell Complex [e.g., *Currie*, 1980; *Mitchell et al*., 1993]. Emplacement of the eastern gabbro and ferroaugite syenites during the first magmatic pulse was followed by intrusion of the remaining rocks of the central and western part. The last magmatic pulse resulted in emplacement of rocks located at the western part of the complex that were magnetized during the interval of normal polarity.

4.4. Implications for the Apparent Reversal Asymmetry and North American Plate Motion

Our new paleomagnetic data indicate that the rocks of Coldwell Complex record a single geomagnetic reversal that is equivalent to the lower reversed-to-normal reversal recorded in the Mamainse Point sequence [*Palmer*, 1970; *Robertson and Fahrig*, 1971; *Swanson-Hysell et al*., 2009] (Figure 9). Although *Swanson-Hysell et al*. [2009] reported a positive reversal test with classification C [*McFadden and McElhinny*, 1990] for this MR1 → MN1 reversal, the test was based on a limited number of paleomagnetic site mean directions ($N = 6$), insufficient to average out the paleosecular variation of geomagnetic field. Moreover, if one of the site mean directions based on only two specimens is excluded, the reversal test becomes indeterminate.

The robust positive reversal test based on a large number of sites presented in this study removes this uncertainty indicating that the reversal was indeed symmetrical. Therefore, our paleomagnetic data provide an additional support to the conclusions reached in prior investigations [e.g., *Davis and Green*, 1997; *Swanson-Hysell et al*., 2009] that the apparent reversal asymmetry is an artifact due to missing paleomagnetic record of an equatorward plate motion in most MCR sequences. On a broader scale, our results support the validity of the geocentric axial dipole assumption for the ~1.1 Ga.

Swanson-Hysell et al. [2009] calculated a superfast rate of 33.6 ± 3.5 cm/yr for the North American apparent polar wander (APW) using the paleomagnetic pole from the Coldwell Complex's Center A [*Lewchuk and Symons*, 1990] and the paleomagnetic pole for the ~1098 Ma normally magnetized flows of the North Shore Volcanics [*Palmer*, 1970; *Books*, 1968]. However, both poles are based on older data sets obtained using outdated techniques.

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We calculated the APW rate based on our new paleomagnetic pole for the Coldwell Complex reversely magnetized rocks (this study) and the paleopole calculated from the high-quality data from the North Shore Volcanics [*Tauxe and Kodama*, 2009]. The new estimate at 24.6 ± 3.8 cm/yr (2.21 ± 0.34°/Myr) is similar to the rate of latitudinal drift of 24.5 ± 3.8 cm/yr calculated from the same data sets. The statistical similarity between the polar wander and latitudinal motion rates indicates that North America experienced latitudinal motion drifting southward with no rotational or longitudinal component in the motion. Although the new estimate is significantly lower than the value from *Swanson-Hysell et al*. [2009], it still exceeds any present-day plate motion rate. However, researchers have long recognized that the present-day velocities may be strongly biased by the peculiarities of present-day plate geometries [*Solomon et al*., 1977] and do not represent an absolute "speed" limit for plates [e.g., *Gordon et al*., 1979; *Tarduno et al*., 1986, 1990; *Meert et al*., 1993]. Our new estimate of the plate velocity for North America at ~1.1 Ga is within the ~15–30 cm/yr range documented for a paleo-Pacific oceanic plate in the Cretaceous [*Tarduno et al*., 1986, 1990] and is not so different from the peak velocity (~18 cm/yr) found for the Indian Plate at ~65–67 Ma [e.g., *Klootwijk et al*., 1992; *Cande et al*., 2010].

Tarduno et al. [1990] proposed that the ultrafast paleo-Pacific plate motion could have been caused by the subduction of old, dense crust along an exceptionally long relative to plate size subducting boundary (i.e., a plate geometry that is not seen in the modern time). *Tarduno et al*. [1990] and [*Tarduno and Myers*, 1994] also suggested that such high velocities might be added by a large-scale reduction of asthenospheric viscosity due to the unusually vigorous mantle plume volcanism in the Pacific basin during the mid-Cretaceous [e.g., *Larson*, 1991]. The effect of mantle plume activity on the plate velocities has been recently revisited by *Cande and Stegman* [2011] who proposed a causative link between the impingement of Reunion plume head and the fast velocity of the Indian Plate between 67 and 52 Ma.

Could one or both of these mechanisms be responsible for the fast plate motion of North America at ~1.1 Ga? The strong slab-pull mechanism is unlikely because Laurentia had been an overriding plate for the most time of its convergence along the Grenville front. Although the polarity of subduction could have changed just before the collision with Amazonia at ~1090 Ma, the crust being subducted was thin and warm [*Hynes and Rivers*, 2010, and references therein] and hence could not provide a sufficiently strong pull for such a large plate.

We note, however, that the period of the fast plate motion of North America between ~1108 and 1098 Ma coincides in time with large-scale intraplate magmatism which, in addition to the ~1110–1087 Ma Midcontinent Rift, led to formation of the 1100–1080 Ma Southwest U.S. Diabase Province [*Hammond*, 1990] and the mafic intrusions in Saskatchewan, ~1400 km northwest of the Midcontinent Rift, with a U-Pb baddeleyite age of 1109 ± 2 Ma [*French et al*., 2002]. These magmatic events likely originate from mantle plume activity [e.g., *Hutchinson et al*., 1990; *Nicholson and Shirey*, 1990]. Nearly coeval with the Laurentian magmatism is the 1112–1103 Ma Umkondo magmatic event represented by sills, dykes, and bimodal volcanics in the Kalahari Craton [*Hanson et al*., 2004]. Although the Umkondo and Keweenawan magmatism were separated by a convergent boundary at that time, they could originate from a single deep-sourced large plume [*Hanson et al*., 2004]. These widespread magmatic events indicate an unusually vigorous mantle beneath Laurentia. We therefore speculate that the fast plate motion may reflect the attendant reduction of asthenospheric drag beneath the plate similar to the mechanism proposed for the mid-Cretaceous Pacific plates by *Tarduno and Myers* [1994].

We note that in contrast to an oceanic plate, the motion of a continental plate such as Laurentia could have been affected by an additional drag caused by the continental root [e.g., *Forsyth and Uyeda*, 1975]. However, prolonged periods of fast plate velocity (in excess of 18 cm/yr) during the Proterozoic were inferred for Laurentia and Gondwana from paleomagnetic analyses [*Meert et al*., 1993]. Our velocity estimate for Laurentia is even higher and provides an additional support to the suggestion that long-term fluctuations in mantle dynamics could alter the balance of forces driving plate motions and allow continental plates to move with the velocities approaching those of the oceanic plates [*Meert et al*., 1993].

As an alternative explanation for the fast equatorward plate motion of Laurentia, a true polar wander (TPW) event was proposed [e.g., *Evans*, 2003]. However, while the observed motion is in principle consistent with this hypothesis, the existence of such a TPW event has not been yet confirmed on a global scale.

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5. Conclusions

Our new paleomagnetic data from the Coldwell Complex indicate that the complex records only a single geomagnetic R → N reversal that is equivalent to the lowermost reversal recorded in the Mamainse Point lava flow sequence. The existence of two reversals suggested in a prior study [*Lewchuk and Symons*, 1990] is likely an artifact due to the deficiencies in their paleomagnetic analyses. The robust positive reversal test obtained in this study indicates that the reversal was symmetrical, hence providing support to the North American Plate motion as the cause of the apparent reversal asymmetry observed in some MCR rocks and discounting the possibility of a long-term nondipolar field at that time. Overall, our results indicate that the geometry and stability of the geomagnetic field at ~1.1 Ga were similar to their Phanerozoic counterparts.

Our paleomagnetic results do not support the complex's emplacement model suggested by *Lewchuk and Symons* [1990] but instead is consistent with earlier models in which the intrusive centers are progressively younger as a result of westward migration of the magma source. The combination of paleomagnetic and radiometric data from other MCR sequences suggests that the entire complex was emplaced within 5–8 Ma which is consistent with prior geochemical and geochronological investigations [e.g., *Heaman and Machado*, 1992]. The new paleomagnetic pole implies an unusually fast plate motion between ~1108 Ma and 1098 Ma, however, at a lower rate than previously suggested by *Swanson-Hysell et al.* [2009]. The fast motion of Laurentia may reflect a decreased mantle drag due to unusually vigorous mantle indicated by widespread intraplate magmatic events. Our results indicate that continental plates can move with the velocities approaching those of the oceanic plates.

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